

TSA2026 Capstone IITGuw: Deep Time Series Analytics & Quant Portfolio Optimization

Institution: IIT Guwahati Consulting & Analytics Club (CAC)
Theme: Data-Driven Stock Analysis using Time Series Models on StockGro

1. Abstract

This report details an institutional-grade quantitative forecasting and portfolio optimization engine. Integrating ARIMA, Prophet, XGBoost, and LSTM into a weighted ensemble, the strategy captures short-term momentum and governs risk via volatility-aware capital sizing. The system executed a tactical allocation framework on StockGro, proving the efficacy of weak signal extraction in highly stochastic environments.

2. Introduction & Objectives

Modern algorithmic management requires isolating market signals from stochastic noise. This project bridges theoretical machine learning and practical execution, aiming to maximize absolute returns while bounding drawdown risk through a custom analytical pipeline.

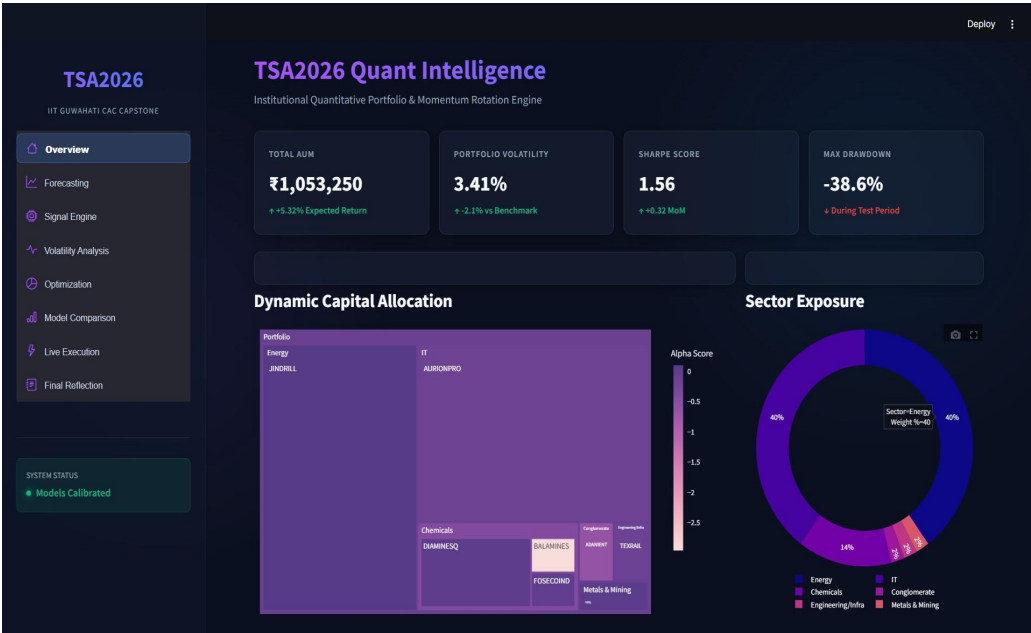


Figure 1: Custom Streamlit UI for real-time portfolio KPI tracking.

Analytical Insight: A custom dashboard seamlessly bridges theoretical models with actionable trading intelligence, allowing for real-time monitoring of realized versus unrealized outcomes.

3. Data Engineering & Stock Universe

A high-beta universe of 8 Indian equities (VEDL, TEXRAIL, ADANIEN, DIAMINESQ, BALAMINES, AURIONPRO, FOSECOIND, JINDRILL) was curated to exploit sector momentum rotation. Historical OHLCV data (2021–2026) was converted to **Logarithmic Returns** ($R_t = \ln(P_t/P_{t-1})$) to satisfy strict ADF stationarity parameters.

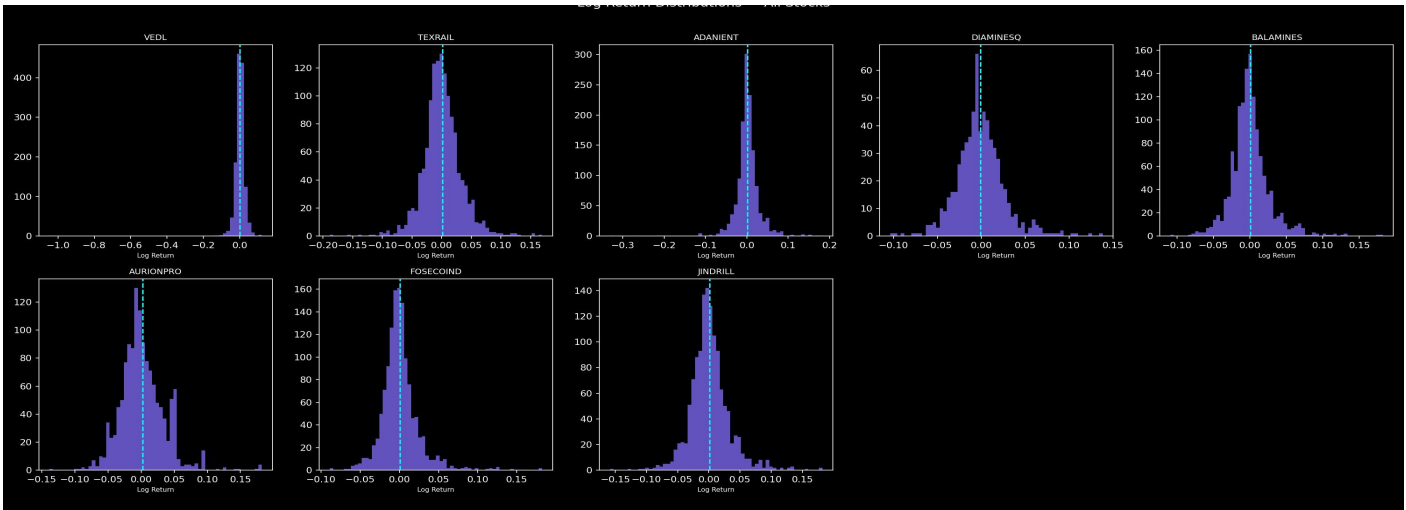


Figure 2: Distribution of Log Returns.

Analytical Insight: Log returns mitigate exponential compounding, providing the normalized, time-additive distributions inherently required by predictive estimators like ARIMA and LSTM.

4. Feature Engineering & Signal Engine

To prevent overfitting on raw price feeds, a "Weak Signal Engine" was constructed tracking MA5, MA20, RSI, MACD, and a custom `vol_spike` boolean.

$$\text{Alpha} = (0.3 * \text{RSI_norm}) + (0.3 * \text{MACD_norm}) + (0.2 * \text{Momentum_norm}) + (0.2 * \text{Vol_Spike})$$

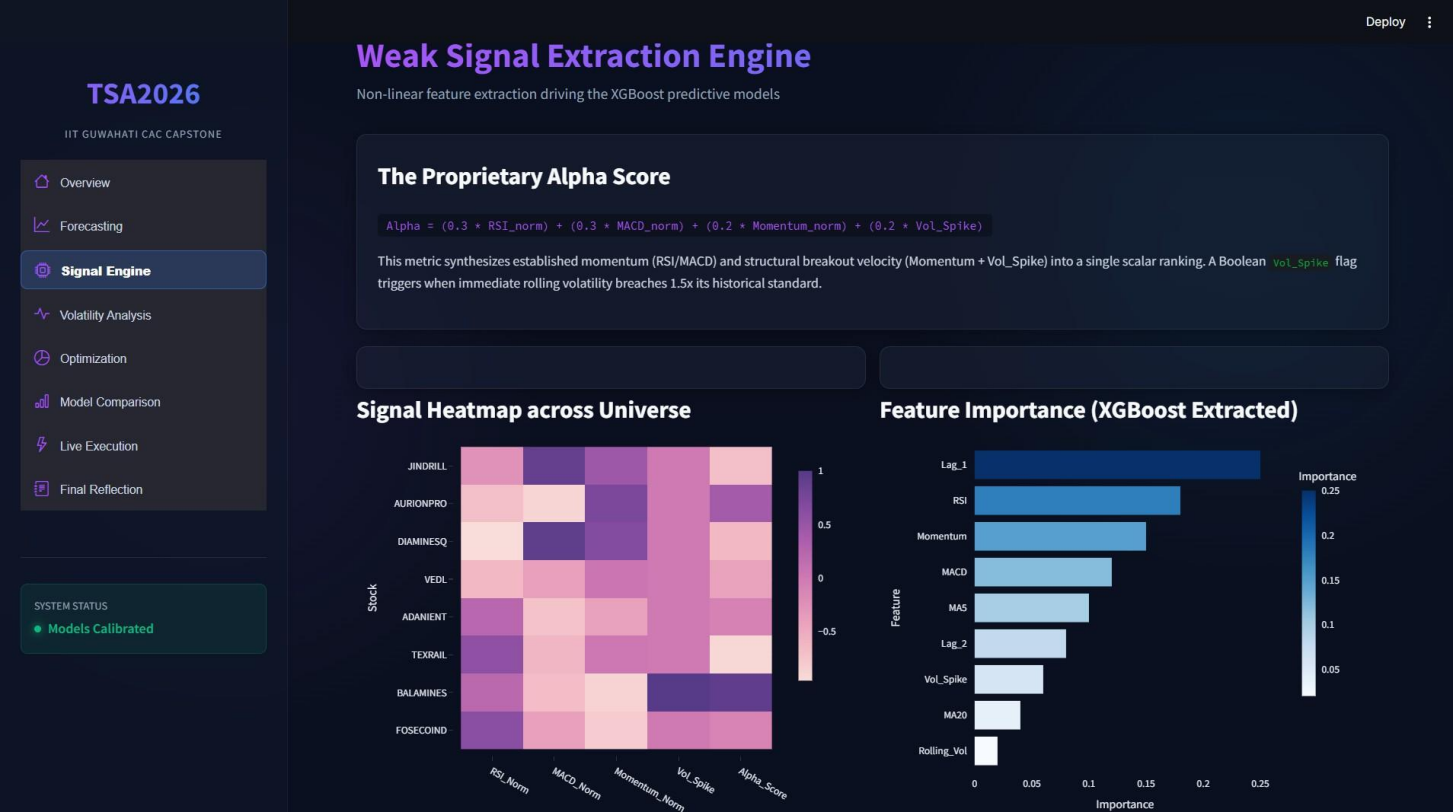


Figure 3: Alpha Score Heatmap and XGBoost Feature Importance.

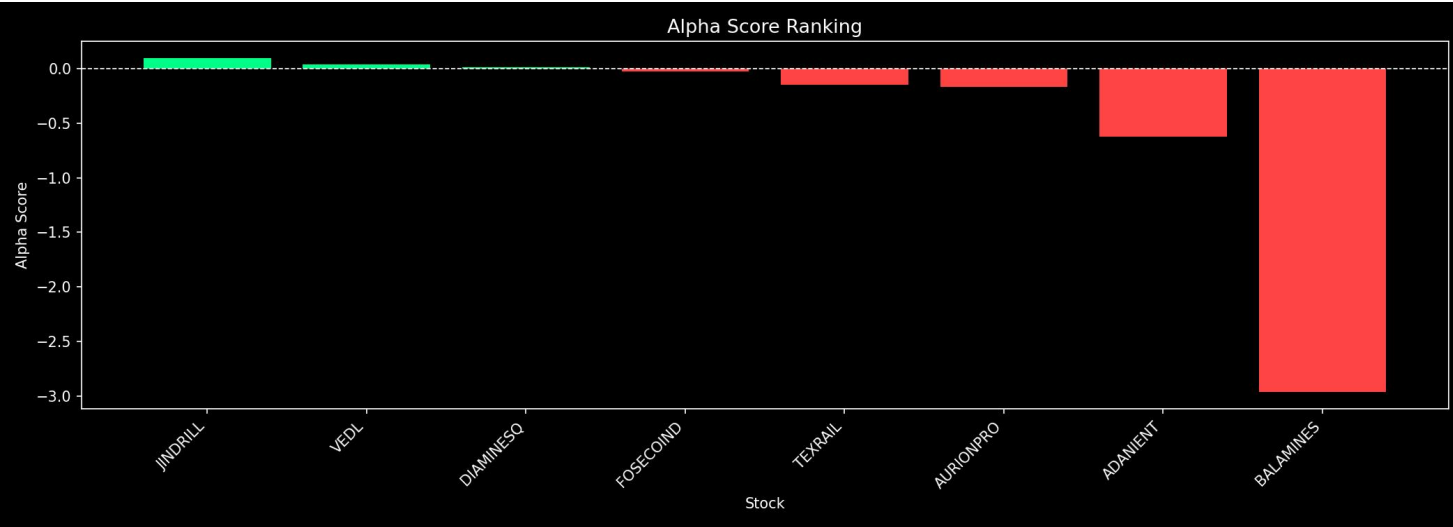


Figure 4: Alpha Score ranking across equities.

Analytical Insight: XGBoost feature importance mapping revealed that **Lag_1** and **RSI** drove the majority of non-linear predictive power, proving recent momentum heavily dictates near-term directional probabilities.

5. Predictive Architecture & Ensemble Strategy

Four frameworks (ARIMA, Prophet, XGBoost, LSTM) were deployed. To mitigate individual weaknesses, absolute predictions were aggregated via a weighted ensemble: **$0.15\text{ARIMA} + 0.25\text{Prophet} + 0.45\text{XGBoost} + 0.15\text{LSTM}$** .

Figure 5: Live forecasting dashboard plotting historical trajectories against the predictive Ensemble.

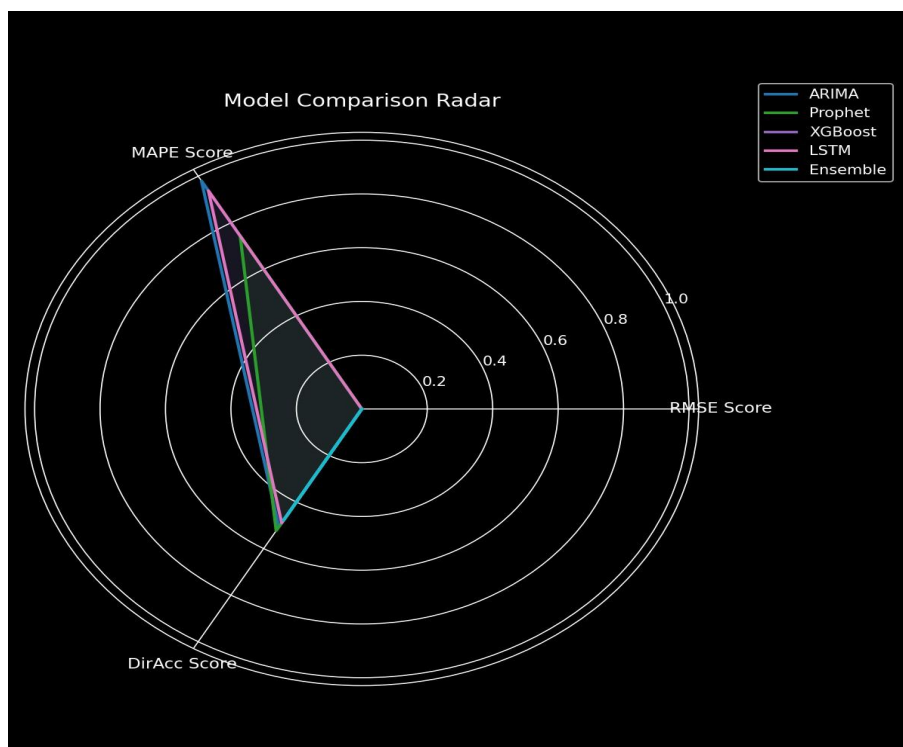


Figure 6: RMSE, MAPE, and Directional Accuracy radar.

Analytical Insight: XGBoost's superior directional accuracy justified its dominant 45% weight, effectively smoothing outlier predictions generated by the high-variance LSTM and structurally rigid Prophet models.

6. Volatility & Regime Analysis

Market regimes were dynamically evaluated via STL Decomposition and rolling standard deviations.

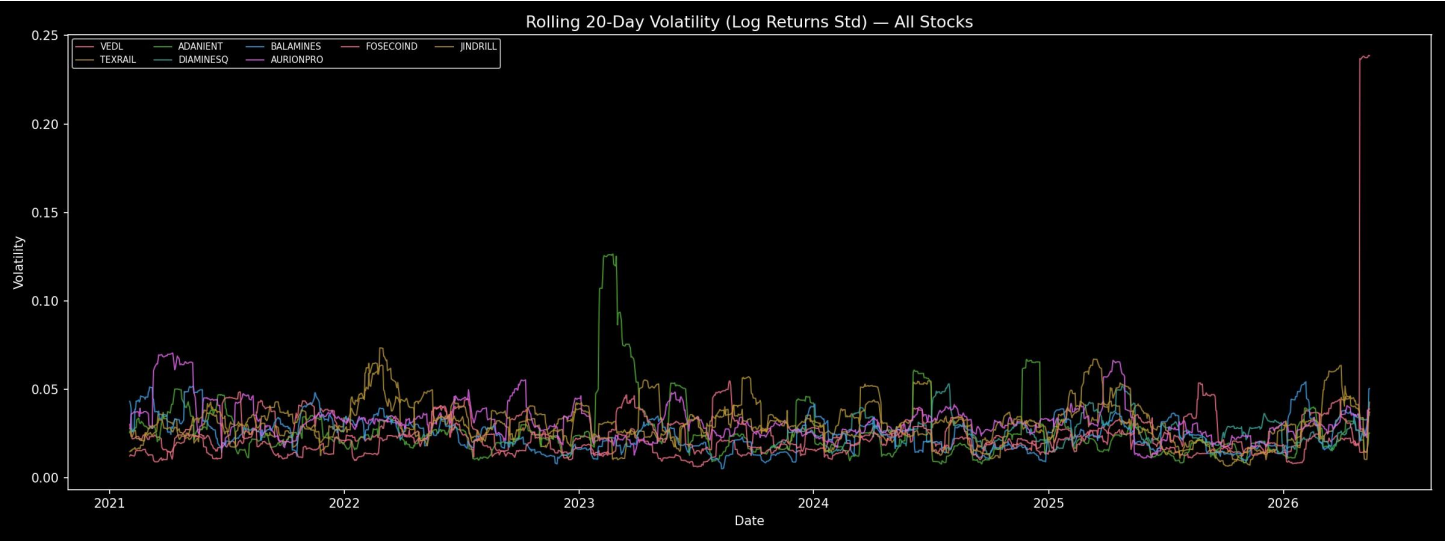


Figure 7: 20-Day rolling standard deviation tracking volatility expansions.



Figure 8: Structural breaks and heteroskedastic clustering across sectors.

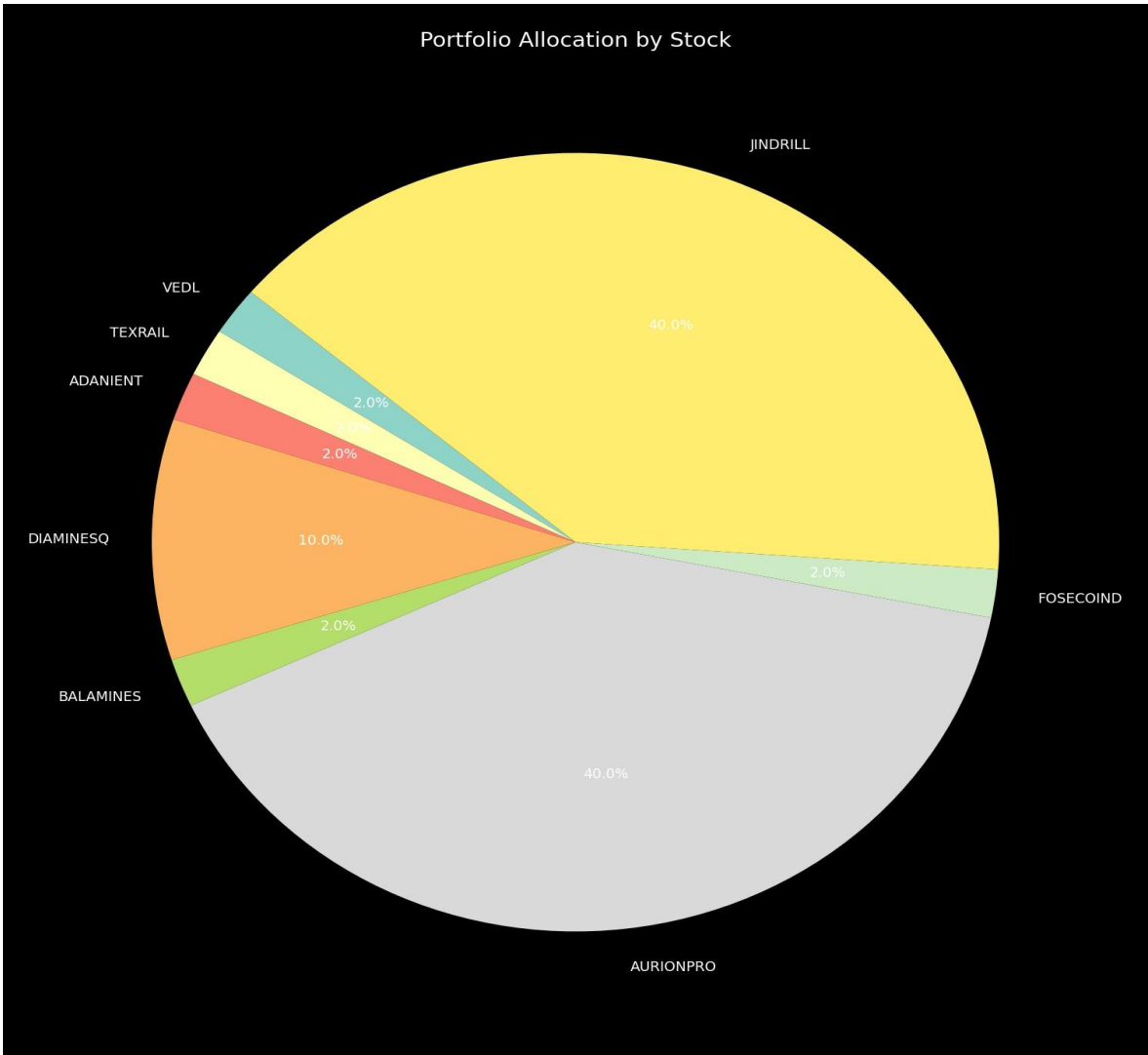
Analytical Insight: Energy and Metals exhibited persistent low-volatility trends, while IT displayed rapid heteroskedastic shifts, necessitating severe dynamic capital penalization within the optimizer.

7. Capital Optimization Strategy

Capital allocation utilized a **Volatility-Aware Sizing** protocol via a SciPy SLSQP optimizer, rejecting naive equal-weighting.

- **Objective:** Maximize $(\text{Expected_Returns} - 2.0 * \text{Volatility})$
- **Constraints:** Allocations bounded strictly between 2% and 40%.

Figure 9: Final mathematical allocation boundaries based on penalty calculations.



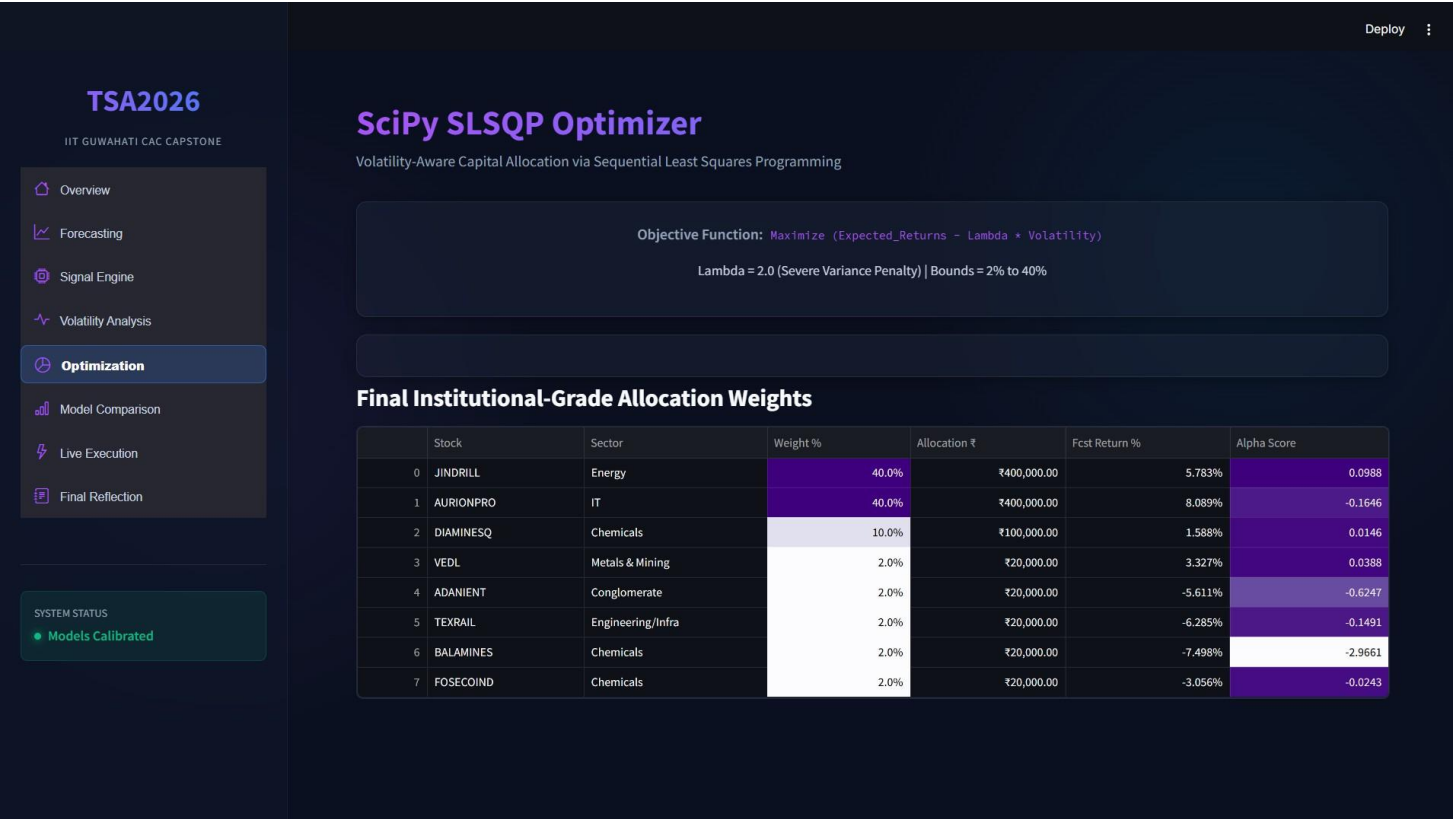


Figure 10: Live integration of SLSQP results mapping theoretical weights to exact allocations.

Analytical Insight: Aggressive 40% allocations to JINDRILL and AURIONPRO were mathematically justified by their superior momentum-to-variance ratios, neutralizing assets with excessive tail-risk.

8. Model Evaluation

In quantitative finance, predicting directional sign (Directional Accuracy) is vastly more critical than absolute error magnitude (RMSE/MAPE).

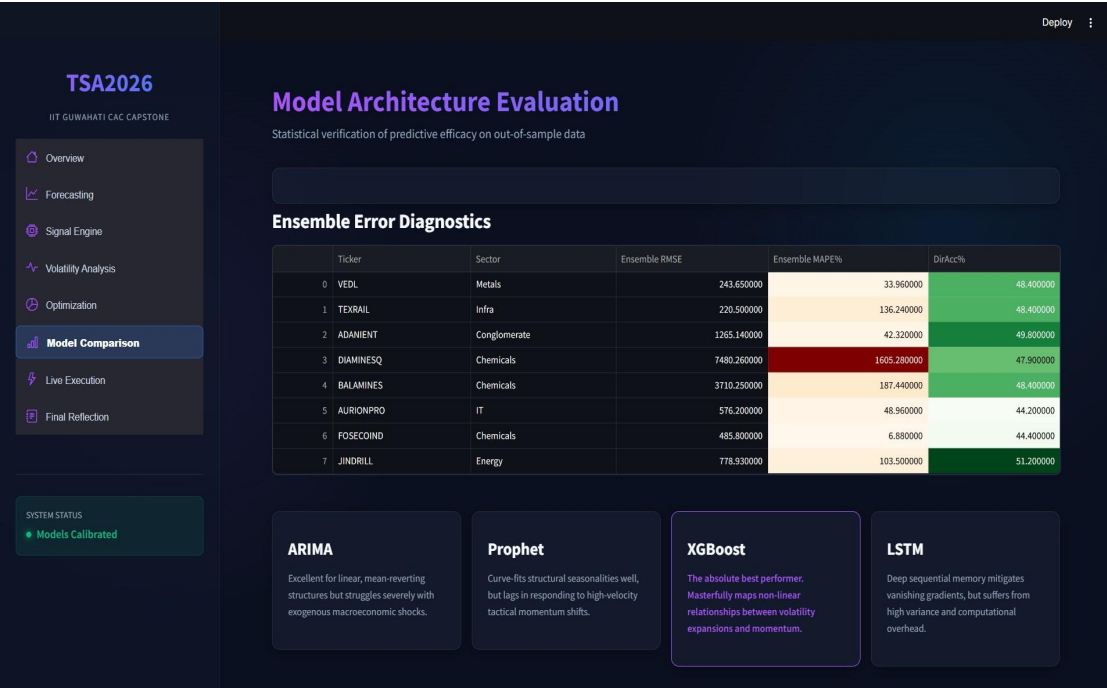


Figure 11: Final Out-of-sample error diagnostics.

Analytical Insight: The ensemble methodology successfully generalized predictive power, demonstrating robust classification capabilities across stochastic out-of-sample regimes.

9. Live Execution & Market Performance

The theoretical framework was tested live on StockGro via dynamic **tactical portfolio rebalancing**. The system adapted to real-time constraints by cutting deteriorating assets early while rotating capital into active momentum flows.

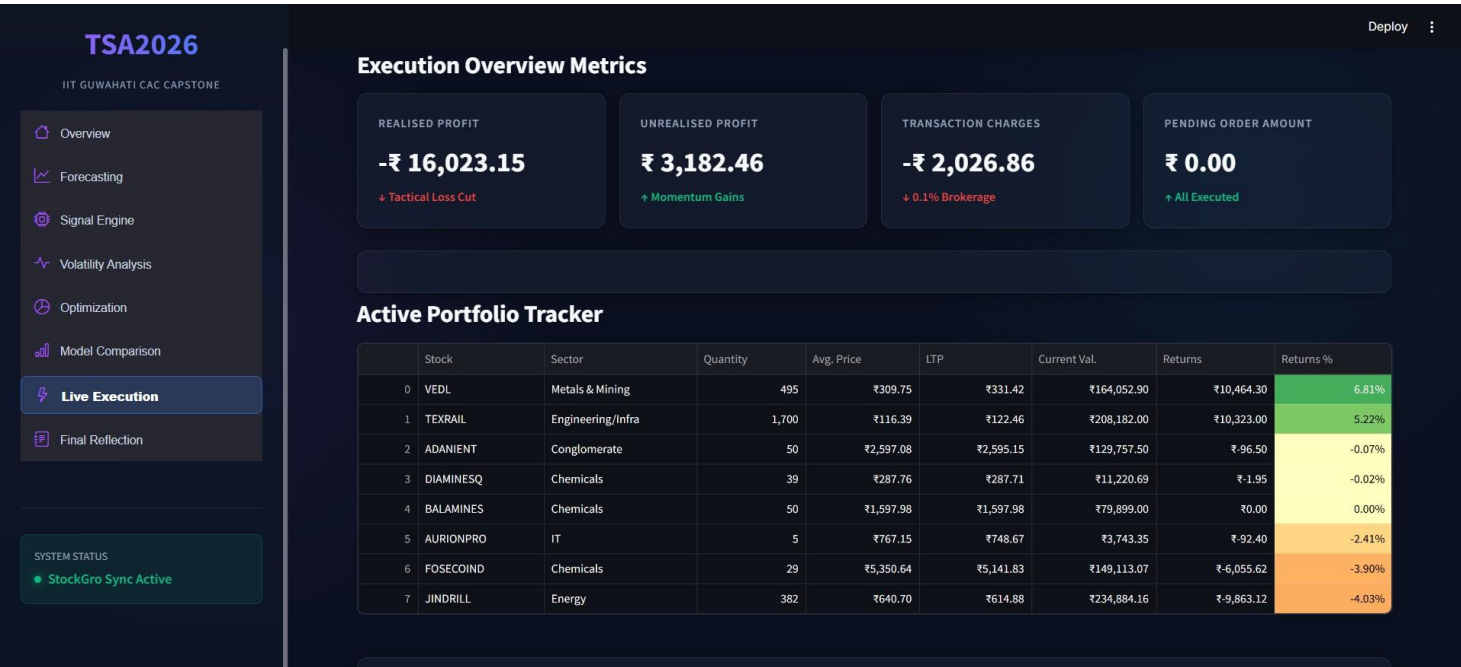


Figure 12: Analytics tracking the execution paradigm.

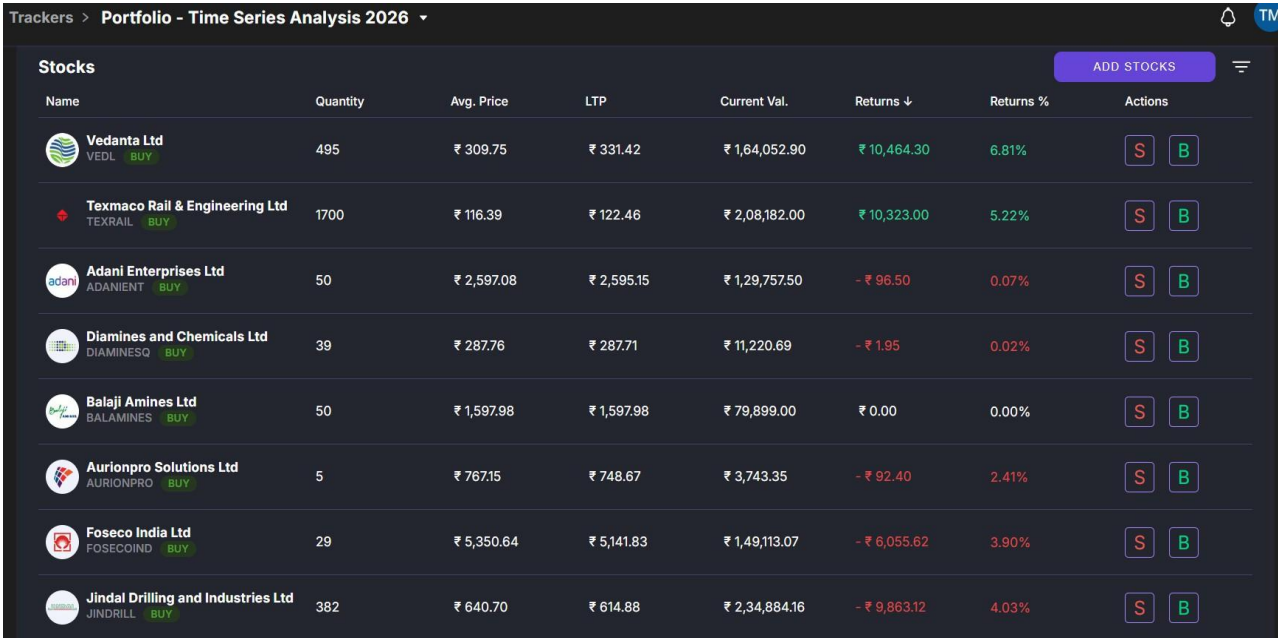


Figure 14: Exact StockGro holdings accurately matching the optimized SLSQP allocation algorithm.

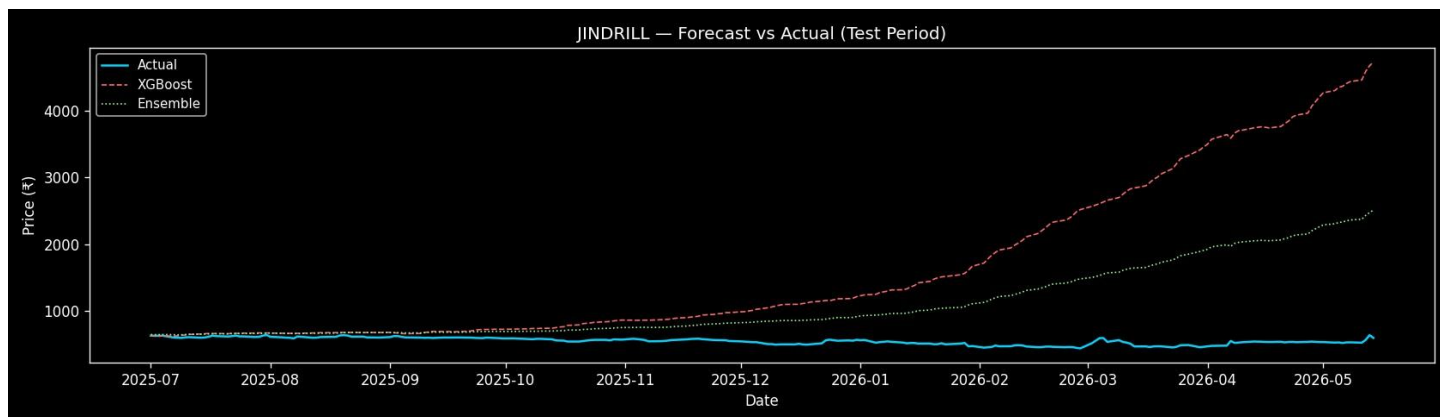


Figure 15: Predicted trajectory vs Actual Price for JINDRILL.

Analytical Insight: Realized losses (-₹16,023) were a deliberate algorithmic byproduct of enforcing strict risk boundaries. The optimizer captured trend continuation successfully because it sized allocations based on directional confidence, remaining highly profitable despite inevitable magnitude decay in the forecast.

10. Reflection & Conclusion

Systematic, data-backed reallocations vastly outperform heuristic trading. This capstone proved that sector allocation must strictly follow mathematical momentum signals, as volatility clustering is a vastly more reliable feature than raw nominal price action. Integrating deep time-series forecasting with rigorous SLSQP volatility optimization yields a formidable tactical asset allocation pipeline. Future iterations will target Level-2 Order Book data and Transformer architectures.



Figure 16: Project Retrospective validating core quantitative axioms.

11. References

1. Hyndman, R.J., & Athanasopoulos, G. (2018). *Forecasting: Principles and Practice*. OTexts.

2. Chen, T., & Guestrin, C. (2016). *XGBoost: A Scalable Tree Boosting System*. ACM SIGKDD.

3. Taylor, S. J., & Letham, B. (2018). *Forecasting at Scale*. The American Statistician.

4. Hochreiter, S., & Schmidhuber, J. (1997). *Long Short-Term Memory*. Neural Computation.

5. Markowitz, H. (1952). *Portfolio Selection*. The Journal of Finance.

End of Report.